

KPI ADVISORY & ANALYTICS

Functional Specification Document

A unified blueprint translating business strategy into governed, measurable performance metrics.

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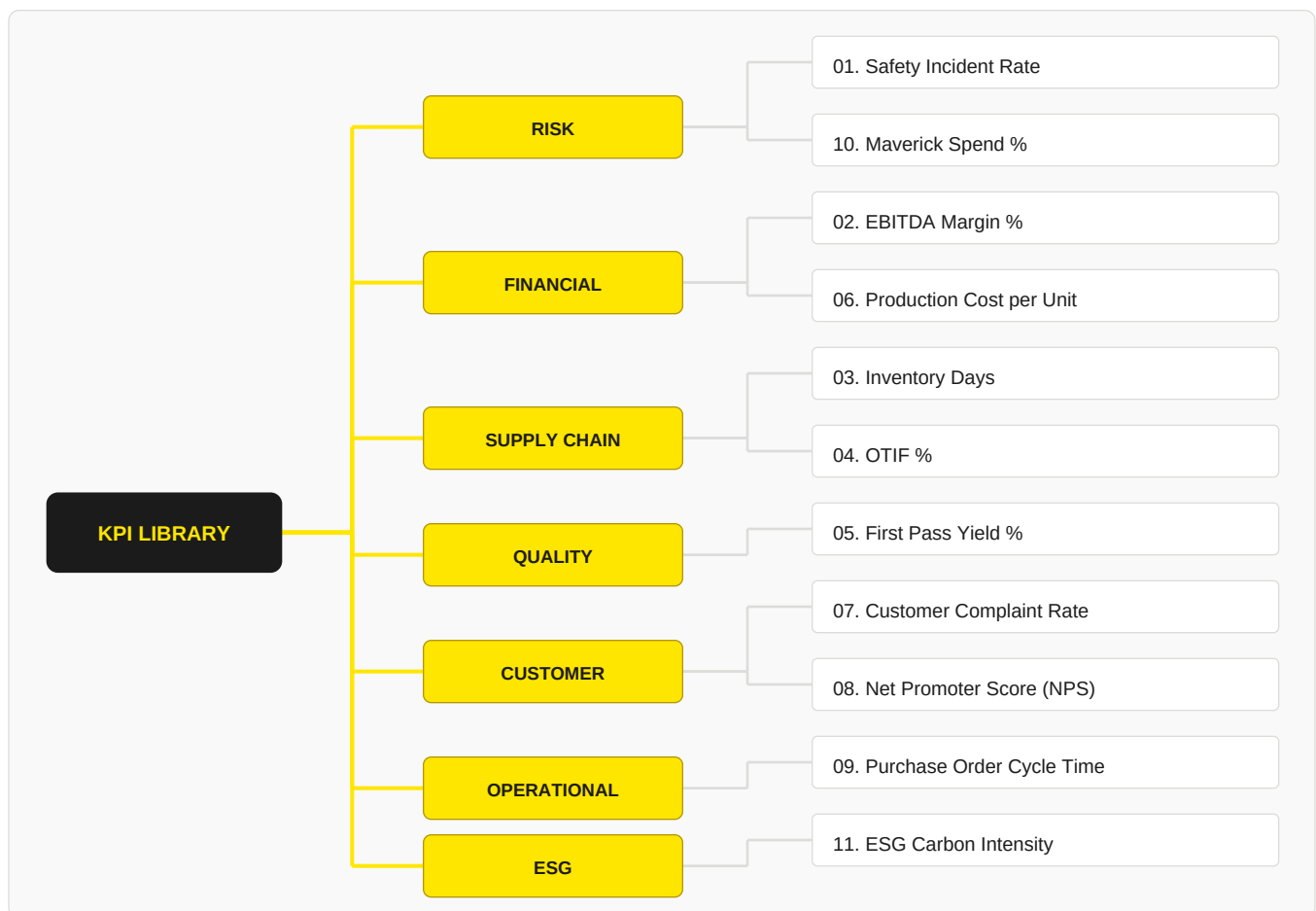
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1. Executive Summary

This Functional Specification Document establishes the governance and calculation standards for the corporate KPI transformation initiative within the Healthcare sector. Guided by executive mandates at the Board level, this initiative directly aligns with key business priorities including Improve Cash Flow, Improve Gross Margin, Enhance Process Automation. In addressing critical challenges such as Margin Leakage from Discounting, Supply Chain Delays, Excess Working Capital Lock-up, the program deploys 11 approved metrics. By detailing data ownership, source systems (SAP/ERP lineage), and calculation boundaries, this document ensures absolute transparency and credibility, providing implementation-ready specifications for engineering and dashboard scripting teams.

2. KPI Landscape Overview

The following visual mind-map displays the KPI Landscape as a branching tree structure, routing from the core KPI Library through strategic categories down to individual approved metrics.



The following table provides a high-level catalog of all approved performance indicators within the scope of this transformation initiative.

KPI ID	KPI Name	Category	Functional Area
KPI-001	Safety Incident Rate	Risk	Operations
KPI-002	EBITDA Margin %	Financial	Finance
KPI-003	Inventory Days	Supply Chain	Supply Chain
KPI-004	OTIF %	Supply Chain	Supply Chain
KPI-005	First Pass Yield %	Quality	Production
KPI-006	Production Cost per Unit	Financial	Production
KPI-007	Customer Complaint Rate	Customer	Quality
KPI-008	Net Promoter Score (NPS)	Customer	Customer Service
KPI-009	Purchase Order Cycle Time	Operational	Procurement
KPI-010	Maverick Spend %	Risk	Procurement
KPI-011	ESG Carbon Intensity	ESG	Operations

3. Strategic Traceability Matrix

This matrix illustrates the strategic alignment from executive objectives down to specific key performance indicators, providing visibility into strategic translation.

Strategic Objective	Business Challenge	KRA	Functional Area	KPI Name
Improve Cash Flow	Margin Leakage from Discounting	Risk Management	Operations	Safety Incident Rate
Improve Cash Flow	Margin Leakage from Discounting	Profitability	Finance	EBITDA Margin %
Improve Cash Flow	Margin Leakage from Discounting	Asset Productivity	Supply Chain	Inventory Days
Improve Cash Flow	Margin Leakage from Discounting	Customer Growth	Supply Chain	OTIF %
Improve Cash Flow	Margin Leakage from Discounting	Operational Excellence	Production	First Pass Yield %
Improve Cash Flow	Margin Leakage from Discounting	Cost Reduction	Production	Production Cost per Unit
Improve Cash Flow	Margin Leakage from Discounting	Customer Growth	Quality	Customer Complaint Rate
Improve Cash Flow	Margin Leakage from Discounting	Customer Growth	Customer Service	Net Promoter Score (NPS)
Improve Cash Flow	Margin Leakage from Discounting	Operational Excellence	Procurement	Purchase Order Cycle Time

Improve Cash Flow	Margin Leakage from Discounting	Cost Reduction	Procurement	Maverick Spend %
Improve Cash Flow	Margin Leakage from Discounting	Risk Management	Operations	ESG Carbon Intensity

4. Individual KPI Specifications

Detailed functional blueprints for each approved metric, outlining definitions, lineage, calculations, and rules.

Metric 01: Safety Incident Rate

KPI Category	Risk	Functional Area	Operations
Business Owner	Head of EHS	Data Owner	Warehouse Safety Coordinator

Business Definition

Measures and evaluates the efficiency of Safety Incident Rate against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Risk Management. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Number of Recordable Safety Incidents} * 200,000) / \text{Total Employee Hours Worked}$
Numerator Details	Number of Recordable Safety Incidents * 200,000
Denominator Details	Total Employee Hours Worked (Employees and Contractors)
Calculation Logic	1. Extract the raw transaction history from SAP EHS (Environment, Health, and Safety). 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP EHS (Environment, Health, and Safety) (Module: EHS). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 02: EBITDA Margin %

KPI Category	Financial	Functional Area	Finance
Business Owner	VP of Finance	Data Owner	Head of Financial Reporting

Business Definition

Measures and evaluates the efficiency of EBITDA Margin % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Profitability. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{EBITDA} / \text{Total Net Revenue}) * 100$
Numerator Details	Earnings before Interest, Tax, Depreciation, and Amortization
Denominator Details	Total Net Revenue
Calculation Logic	1. Extract the raw transaction history from SAP General Ledger. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then $\text{KPI} = (9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP General Ledger (Module: FI). Dependencies: Master transactional tables and inventory logs.

Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 03: Inventory Days

KPI Category	Supply Chain	Functional Area	Supply Chain
Business Owner	Director of Supply Chain	Data Owner	Inventory Control Manager

Business Definition

Measures and evaluates the efficiency of Inventory Days against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Asset Productivity. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Average Inventory Value} / \text{Annual Cost of Goods Sold}) * 365$
Numerator Details	Average Inventory Value
Denominator Details	Annual Cost of Goods Sold (COGS)
Calculation Logic	1. Extract the raw transaction history from SAP MM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP MM (Module: MM). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 04: OTIF %

KPI Category	Supply Chain	Functional Area	Supply Chain
Business Owner	VP of Logistics	Data Owner	Logistics Operations Manager

Business Definition

Measures and evaluates the efficiency of OTIF % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Customer Growth. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Orders Delivered On-Time and In-Full} / \text{Total Orders Shipped}) * 100$
Numerator Details	Orders Delivered On-Time and In-Full
Denominator Details	Total Orders Shipped
Calculation Logic	1. Extract the raw transaction history from SAP SD / EWM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then $KPI = (9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP SD / EWM (Module: SD/MM). Dependencies: Master transactional tables and inventory logs.

Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 05: First Pass Yield %

KPI Category	Quality	Functional Area	Production
Business Owner	Director of Manufacturing	Data Owner	Quality Control Manager

Business Definition

Measures and evaluates the efficiency of First Pass Yield % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Operational Excellence. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Good Units Produced with No Rework} / \text{Total Units Started}) * 100$
Numerator Details	Good Units Produced with No Rework
Denominator Details	Total Units Started
Calculation Logic	1. Extract the raw transaction history from SAP QM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then $KPI = (9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP QM (Module: PP/QM). Dependencies: Master transactional tables and inventory logs.

Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 06: Production Cost per Unit

KPI Category	Financial	Functional Area	Production
Business Owner	VP of Operations	Data Owner	Cost Accounting Manager

Business Definition

Measures and evaluates the efficiency of Production Cost per Unit against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Cost Reduction. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	Total Manufacturing Cost in Period / Total Good Units Produced
Numerator Details	Direct Materials + Direct Labor + Applied Overhead
Denominator Details	Total Good Units Produced
Calculation Logic	1. Extract the raw transaction history from SAP CO (Product Cost Controlling). 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP CO (Product Cost Controlling) (Module: PP/CO). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 07: Customer Complaint Rate

KPI Category	Customer	Functional Area	Quality
Business Owner	Chief Customer Officer	Data Owner	Customer Service Manager

Business Definition

Measures and evaluates the efficiency of Customer Complaint Rate against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Customer Growth. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Customer Complaints Received} / \text{Total Orders Shipped}) * 100$
Numerator Details	Customer Complaints Received
Denominator Details	Total Orders Shipped
Calculation Logic	1. Extract the raw transaction history from Salesforce CRM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: Salesforce CRM (Module: QM). Dependencies: Master transactional tables and inventory logs.
Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 08: Net Promoter Score (NPS)

KPI Category	Customer	Functional Area	Customer Service
Business Owner	Chief Marketing Officer	Data Owner	Market Research Manager

Business Definition

Measures and evaluates the efficiency of Net Promoter Score (NPS) against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Customer Growth. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	% Promoters (score 9-10) - % Detractors (score 0-6)
Numerator Details	Promoters (score 9-10) - Detractors (score 0-6)
Denominator Details	Total Survey Responders
Calculation Logic	1. Extract the raw transaction history from Qualtrics CX Platform. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: Qualtrics CX Platform (Module: SD). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 09: Purchase Order Cycle Time

KPI Category	Operational	Functional Area	Procurement
Business Owner	Chief Procurement Officer	Data Owner	Procurement Systems Administrator

Business Definition

Measures and evaluates the efficiency of Purchase Order Cycle Time against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Operational Excellence. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	Sum of (PO Dispatch Time - PR Submission Time) / Total POs Dispatched
Numerator Details	Sum of Hours from PR Submission to PO Dispatch
Denominator Details	Total POs Dispatched
Calculation Logic	1. Extract the raw transaction history from SAP ERP (MM Module). 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = (9,500 / 10,000) * 100 = 95.00%.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP ERP (MM Module) (Module: MM). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 10: Maverick Spend %

KPI Category	Risk	Functional Area	Procurement
Business Owner	Chief Procurement Officer	Data Owner	Spend Analysis Manager

Business Definition

Measures and evaluates the efficiency of Maverick Spend % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Cost Reduction. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Procurement Value Outside Contract} / \text{Total Procurement Spend}) * 100$
Numerator Details	Procurement Value Outside Contract
Denominator Details	Total Procurement Spend
Calculation Logic	1. Extract the raw transaction history from SAP Ariba. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP Ariba (Module: MM). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

4. Individual KPI Specifications (Continued)

Metric 11: ESG Carbon Intensity

KPI Category	ESG	Functional Area	Operations
Business Owner	Chief Sustainability Officer	Data Owner	Environmental Compliance Manager

Business Definition

Measures and evaluates the efficiency of ESG Carbon Intensity against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Risk Management. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	Metric Tons of CO2 equivalent (Scope 1 & 2) / Total Production Units
Numerator Details	Metric Tons of CO2 equivalent (Scope 1 & 2)
Denominator Details	Total Production Units
Calculation Logic	1. Extract the raw transaction history from SAP Sustainability Control Tower. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = (9,500 / 10,000) * 100 = 95.00%.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP Sustainability Control Tower (Module: ESG). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

5. Governance Framework

To ensure metric consistency, accountability, and ongoing relevance, a formal governance framework is established for all approved KPIs. This structure assigns clear responsibilities and defines escalation paths.

Roles and Responsibilities

- **Business Owner:** Responsible for defining the business logic, validating calculation results, approving target thresholds, and driving operational performance based on metric insights.
- **Data Owner:** Accountable for technical lineage, data completeness, source-to-target mapping, ETL data quality checks, and resolving data ingestion or availability issues.
- **Escalation Path:** In case of data quality discrepancies or alignment disputes, issues are escalated to the Data Governance Committee and KPI Advisory Board for review and reconciliation.

Ownership and Governance Policy Matrix

Metric Name	Business Owner	Data Owner	Governance Policy Notes
Safety Incident Rate	Head of EHS	Warehouse Safety Coordinator	Business Owner (Head of EHS): Reviews variance and coordinates corrective actions. Data Owner (Warehouse Safety Coordinator): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
EBITDA Margin %	VP of Finance	Head of Financial Reporting	Business Owner (VP of Finance): Reviews variance and coordinates corrective actions. Data Owner (Head of Financial Reporting): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Inventory Days	Director of Supply Chain	Inventory Control Manager	Business Owner (Director of Supply Chain): Reviews variance and coordinates corrective actions. Data Owner (Inventory Control Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
OTIF %	VP of Logistics	Logistics Operations Manager	Business Owner (VP of Logistics): Reviews variance and coordinates corrective actions. Data Owner (Logistics Operations Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
First Pass Yield %	Director of Manufacturing	Quality Control Manager	Business Owner (Director of Manufacturing): Reviews variance and coordinates corrective actions. Data Owner (Quality Control Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.

Production Cost per Unit	VP of Operations	Cost Accounting Manager	Business Owner (VP of Operations): Reviews variance and coordinates corrective actions. Data Owner (Cost Accounting Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Customer Complaint Rate	Chief Customer Officer	Customer Service Manager	Business Owner (Chief Customer Officer): Reviews variance and coordinates corrective actions. Data Owner (Customer Service Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Net Promoter Score (NPS)	Chief Marketing Officer	Market Research Manager	Business Owner (Chief Marketing Officer): Reviews variance and coordinates corrective actions. Data Owner (Market Research Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Purchase Order Cycle Time	Chief Procurement Officer	Procurement Systems Administrator	Business Owner (Chief Procurement Officer): Reviews variance and coordinates corrective actions. Data Owner (Procurement Systems Administrator): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Maverick Spend %	Chief Procurement Officer	Spend Analysis Manager	Business Owner (Chief Procurement Officer): Reviews variance and coordinates corrective actions. Data Owner (Spend Analysis Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
ESG Carbon Intensity	Chief Sustainability Officer	Environmental Compliance Manager	Business Owner (Chief Sustainability Officer): Reviews variance and coordinates corrective actions. Data Owner (Environmental Compliance Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.

6. Reporting & Dashboard Requirements

Visualization layout and threshold criteria dictate how data is displayed to support decision-making. The matrix below outlines reporting recommendations and performance thresholds for each metric.

Metric Name	Reporting Guidelines	Dashboard Placement	Threshold Guidance
Safety Incident Rate	Visual layout: Gauge card and trend line showing actuals vs target of < 1.5.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red > 2.5, Amber 1.5-2.5, Green < 1.5).
EBITDA Margin %	Visual layout: Gauge card and trend line showing actuals vs target of 15-20%.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red < 15%, Amber 15-17%, Green >= 17%).
Inventory Days	Visual layout: Gauge card and trend line showing actuals vs target of 30-60 days.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red > 75 days, Amber 60-75 days, Green 30-60 days).
OTIF %	Visual layout: Gauge card and trend line showing actuals vs target of >= 95%.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red < 90%, Amber 90-95%, Green >= 95%).
First Pass Yield %	Visual layout: Gauge card and trend line showing actuals vs target of >= 98%.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red < 95%, Amber 95-98%, Green >= 98%).
Production Cost per Unit	Visual layout: Gauge card and trend line showing actuals vs target of < \$50.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red > \$60, Amber \$50-\$60, Green < \$50).
Customer Complaint Rate	Visual layout: Gauge card and trend line showing actuals vs target of < 2 per 1000.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red > 4 per 1000, Amber 2-4 per 1000, Green < 2 per 1000).
Net Promoter Score (NPS)	Visual layout: Gauge card and trend line showing actuals vs target of > 50.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red < 30, Amber 30-50, Green > 50).
Purchase Order Cycle Time	Visual layout: Gauge card and trend line showing actuals vs target of < 3 business days.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red > 5 days, Amber 3-5 days, Green < 3 days).
Maverick Spend %	Visual layout: Gauge card and trend line showing actuals vs target of < 5%.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red > 10%, Amber 5-10%, Green < 5%).
ESG Carbon Intensity	Visual layout: Gauge card and trend line showing actuals vs target of < 0.1 kg CO2e / \$ Revenue.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (Red > 0.15, Amber 0.10-0.15, Green < 0.10).

7. Assumptions & Constraints

A clear understanding of business and technical assumptions is critical for successful implementation. The following list represents consolidated baseline assumptions and constraints.

General Architectural Assumptions

- **Data Availability:** Source transactional tables are assumed to be loaded into the central reporting repository on a standard nightly batch cadence.
- **Fiscal Calendar:** Standard calendar year rules are assumed unless otherwise explicitly documented in specific financial indicators.
- **System Uptime:** Target source ERP ledgers are assumed to maintain 99.5% uptime during standard reporting extraction windows.

Metric-Specific Assumptions and Limitations

- **Safety Incident Rate:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **EBITDA Margin %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Inventory Days:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **OTIF %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **First Pass Yield %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Production Cost per Unit:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Customer Complaint Rate:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Net Promoter Score (NPS):** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Purchase Order Cycle Time:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Maverick Spend %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **ESG Carbon Intensity:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.

8. Implementation Considerations

Transitioning these specifications into functional BI tools requires rigorous testing, change management, and data reconciliation. Standard implementation considerations are outlined below.

Standard Implementation Guidelines

- **Data Reconciliation:** All KPI calculation outcomes must be audited and reconciled against official books of record or audited financial statements.
- **User Acceptance Testing (UAT):** Business owners must perform visual and numeric validation of dashboard mockups prior to production sign-off.
- **Change Management:** Training sessions and clear system documentation are required to support user onboarding and ensure high organizational adoption.

Metric-Specific Technical Considerations

- **Safety Incident Rate:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **EBITDA Margin %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Inventory Days:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **OTIF %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **First Pass Yield %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Production Cost per Unit:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Customer Complaint Rate:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Net Promoter Score (NPS):** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Purchase Order Cycle Time:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Maverick Spend %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **ESG Carbon Intensity:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.

9. Appendix

KPI Glossary

- **KPI (Key Performance Indicator):** A quantifiable measure used to evaluate the success of an organization or activity in meeting performance objectives.
- **KRA (Key Result Area):** Primary focus areas of outcomes or outputs for which an organizational unit or role is responsible.
- **Numerator:** The upper portion of a division representing the measured subset of occurrences.
- **Denominator:** The lower portion of a division representing the total base population.

Data Quality Principles

- **Accuracy:** Data correctly represents the real-world operational event it records.
- **Completeness:** All necessary dataset components are present without omission.
- **Consistency:** Metrics align across various systems, business units, and report interfaces.
- **Timeliness:** Updates occur within the required reporting cadence and operational windows.

Acronym Reference

- **ERP:** Enterprise Resource Planning
- **SAP FI-CO:** Financial Accounting & Controlling
- **SAP SD:** Sales & Distribution
- **SAP MM:** Materials Management
- **BI / DWH:** Business Intelligence / Data Warehouse
- **UAT:** User Acceptance Testing